

SF1811 Optimization

Lecture 14: Nonlinear least-squares problems and the Gauss–Newton method

Chapter 17

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Nonlinear least squares and Gauss–Newton

Main question:

How can a nonlinear least-squares problem be solved by a sequence of linear least-squares problems?

Today:

- Nonlinear residuals and least-squares objectives.
- Gradient and Hessian structure.
- Linearization of the residual vector.
- The Gauss–Newton direction.
- Step lengths and possible difficulties.
- Nonlinear curve fitting.

Nonlinear least-squares problems

Let

$$h : \mathbb{R}^n \to \mathbb{R}^m, \qquad h(x) = \begin{pmatrix} h_1(x) \\ \vdots \\ h_m(x) \end{pmatrix}.$$

A nonlinear least-squares problem has the form

$$(NLS) \qquad \underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x) = \frac{1}{2}\|h(x)\|^2.$$

Equivalently,

$$f(x) = \frac{1}{2}\sum_{i=1}^{m} h_i(x)^2.$$

The functions $h_i(x)$ are called the residual functions.

The objective is nonlinear, but retains a sum-of-squares structure.

Nonlinear model fitting

Suppose that measurements

$$(t_1, s_1), \ldots, (t_m, s_m)$$

are approximated by a nonlinear model

$$g(t; x), \qquad x \in \mathbb{R}^n.$$

Define the residuals $h_i(x) = g(t_i; x) - s_i$, $i = 1, \ldots, m$.

The model parameters are chosen by solving

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2} \sum_{i=1}^{m} \left(g(t_i; x) - s_i \right)^2 .$$

Examples of nonlinear models include

$$g(t; x) = x_1 e^{x_2 t} \quad \text{and} \quad g(t; x) = \frac{x_1 t}{x_2 + t}.$$

The parameters enter the model nonlinearly.

The residual Jacobian

Let

$$J(x) = \nabla h(x) \in \mathbb{R}^{m \times n}$$

denote the Jacobian of the residual vector, with entries

$$J_{ij}(x) = \frac{\partial h_i(x)}{\partial x_j}.$$

Thus the ith row of $J(x)$ is

$$\nabla h_i(x).$$

For a step $d \in \mathbb{R}^n$,

$$h(x + d) = h(x) + J(x)d + o(\|d\|).$$

The Jacobian describes the first-order change in all residuals.

Gradient of the least-squares objective

Since

$$f(x) = \frac{1}{2} h(x)^{\mathsf{T}} h(x),$$

the chain rule gives

$$\nabla f(x) = h(x)^{\mathsf{T}} J(x).$$

Equivalently,

$$\nabla f(x)^{\mathsf{T}} = J(x)^{\mathsf{T}} h(x).$$

Componentwise,

$$\nabla f(x) = \sum_{i=1}^{m} h_i(x) \nabla h_i(x).$$

Therefore, the first-order necessary condition is $J(\hat{x})^{\mathsf{T}} h(\hat{x}) = 0$.

At a stationary point, the residual vector is orthogonal to every first-order model direction.

The exact Hessian

Let $H_i(x)$ denote the Hessian of the residual function h_i. Differentiating the gradient gives

$$\boxed{F(x) = J(x)^{\mathsf{T}} J(x) + \sum_{i=1}^{m} h_i(x) H_i(x).}$$

The Hessian has two parts:

$$J(x)^{\mathsf{T}} J(x) \succeq 0$$

comes from the first derivatives of the residuals, while

$$\sum_{i=1}^{m} h_i(x) H_i(x)$$

contains the second derivatives of the residuals.

If the residuals are linear functions, then $H_i(x) = 0$, $i = 1, \ldots, m$, and the second term vanishes.

Newton's equation for nonlinear least squares

Newton's method solves

$$F(x^{(k)})d^{(k)} = -\nabla f(x^{(k)})^{\mathsf{T}}.$$

For a nonlinear least-squares objective, this becomes

$$\left(J_k^{\mathsf{T}} J_k + \sum_{i=1}^{m} h_i(x^{(k)}) H_i(x^{(k)})\right) d^{(k)} = -J_k^{\mathsf{T}} h_k,$$

where

$$J_k = J(x^{(k)}), \qquad h_k = h(x^{(k)}).$$

This requires both first and second derivatives of every residual function.

Can the sum-of-squares structure provide a simpler model?

Linearizing the residual vector

At the current iterate $x^{(k)}$, use the approximation

$$h(x^{(k)} + d) \approx h_k + J_k d.$$

Substituting this into the least-squares objective gives the local model

$$q_k(d) = \frac{1}{2}\|h_k + J_k d\|^2.$$

Thus the step is determined by the linear least-squares problem

$$\boxed{\underset{d \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2}\|J_k d + h_k\|^2.}$$

Each nonlinear iteration is reduced to the problem studied in L10.

The Gauss–Newton equation

The normal equations for

$$\underset{d}{\text{minimize}} \quad \frac{1}{2}\|J_k d + h_k\|^2$$

are

$$J_k^{\mathsf{T}} J_k d = -J_k^{\mathsf{T}} h_k.$$

The solution $d^{(k)}$ is the Gauss–Newton direction:

$$\boxed{J(x^{(k)})^{\mathsf{T}} J(x^{(k)}) d^{(k)} = -J(x^{(k)})^{\mathsf{T}} h(x^{(k)}).}$$

If J_k has full column rank, then

$$J_k^{\mathsf{T}} J_k \succ 0,$$

and the Gauss–Newton direction is uniquely determined.

Gauss–Newton as an approximate Newton method

The exact Newton matrix is

$$F(x) = J(x)^{\mathsf{T}} J(x) + \sum_{i=1}^{m} h_i(x) H_i(x).$$

Gauss–Newton uses the approximation

$$F(x) \approx J(x)^{\mathsf{T}} J(x).$$

Consequently:

$$\begin{aligned} \text{Newton:} \quad & F(x^{(k)}) d^{(k)} && = -J_k^{\mathsf{T}} h_k, \\ \text{Gauss–Newton:} \quad & J_k^{\mathsf{T}} J_k d^{(k)} && = -J_k^{\mathsf{T}} h_k. \end{aligned}$$

Gauss–Newton neglects the residual-weighted second derivatives.

When is the approximation accurate?

The omitted term is

$$\sum_{i=1}^{m} h_i(x) H_i(x).$$

The Gauss–Newton approximation is therefore expected to be good when:

- the residuals $h_i(x)$ are small near the solution;
- the residual functions are nearly linear near the solution;
- the second derivatives $H_i(x)$ are moderate.

If

$$h(\hat{x}) = 0,$$

then at the solution

$$F(\hat{x}) = J(\hat{x})^{\mathsf{T}} J(\hat{x}).$$

Thus Gauss–Newton uses the exact Hessian at a zero-residual solution.

Small residuals make Gauss–Newton resemble Newton's method.

The Gauss–Newton direction is a descent direction

The Gauss–Newton equation is

$$J_k^{\mathsf{T}} J_k d^{(k)} = -J_k^{\mathsf{T}} h_k.$$

Multiplying from the left by $d^{(k)^{\mathsf{T}}}$ gives

$$d^{(k)^{\mathsf{T}}} J_k^{\mathsf{T}} h_k = -d^{(k)^{\mathsf{T}}} J_k^{\mathsf{T}} J_k d^{(k)}.$$

Since

$$\nabla f(x^{(k)}) = h_k^{\mathsf{T}} J_k,$$

we obtain

$$\nabla f(x^{(k)}) d^{(k)} = -\|J_k d^{(k)}\|^2.$$

If J_k has full column rank and $d^{(k)} \neq 0$, then

$$\nabla f(x^{(k)}) d^{(k)} < 0.$$

Thus the Gauss–Newton direction is then a descent direction.

The Gauss–Newton method

Given an initial point $x^{(0)}$, repeat:

1. Evaluate the residual vector and its Jacobian:

$$h_k = h(x^{(k)}), \qquad J_k = J(x^{(k)}).$$

2. Solve the linear least-squares problem

$$\underset{d}{\text{minimize}} \quad \frac{1}{2}\|J_k d + h_k\|^2.$$

3. Choose a step length $t_k > 0$.
4. Update

$$x^{(k+1)} = x^{(k)} + t_k d^{(k)}.$$

Try a full step $t_k = 1$ first. If it does not reduce the objective, use a shorter step.

Full and shorter steps

The Gauss–Newton model predicts

$$h(x^{(k)} + d^{(k)}) \approx h_k + J_k d^{(k)}.$$

A full step is appropriate when this linearization is accurate over the entire step.

Farther from a solution, the linearization may be poor. Then choose

$$x^{(k+1)} = x^{(k)} + t_k d^{(k)}, \qquad 0 < t_k < 1,$$

so that

$$f(x^{(k+1)}) < f(x^{(k)}).$$

The direction comes from the model; the step length controls trust in the model.

A nonlinear curve-fitting example

Suppose that the model is

$$g(t; x) = x_1 e^{x_2 t}$$

and the measurements are

$$(t_1, s_1), \ldots, (t_m, s_m).$$

The residual functions are

$$h_i(x) = x_1 e^{x_2 t_i} - s_i.$$

Their derivatives are

$$\frac{\partial h_i(x)}{\partial x_1} = e^{x_2 t_i},$$

and

$$\frac{\partial h_i(x)}{\partial x_2} = x_1 t_i e^{x_2 t_i}.$$

Thus row i of the Jacobian is

$$\begin{pmatrix} e^{x_2 t_i} & x_1 t_i e^{x_2 t_i} \end{pmatrix}.$$

One Gauss–Newton iteration

At the current iterate $x^{(k)}$, form

$$h_k = \begin{pmatrix} h_1(x^{(k)}) \\ \vdots \\ h_m(x^{(k)}) \end{pmatrix} \quad \text{and} \quad J_k = \begin{pmatrix} \nabla h_1(x^{(k)}) \\ \vdots \\ \nabla h_m(x^{(k)}) \end{pmatrix}.$$

Then solve

$$\underset{d \in \mathbb{R}^2}{\text{minimize}} \quad \frac{1}{2}\|J_k d + h_k\|^2.$$

The updated parameters are

$$x^{(k+1)} = x^{(k)} + t_k d^{(k)}.$$

The linearized fitted values satisfy

$$g(t_i; x^{(k)} + d) \approx g(t_i; x^{(k)}) + \nabla_x g(t_i; x^{(k)}) d.$$

Linear least squares is a special case

Suppose that

$$h(x) = Ax - b.$$

Then

$$J(x) = A$$

is constant and

$$H_i(x) = 0, \qquad i = 1, \ldots, m.$$

The Gauss–Newton subproblem is

$$\underset{d}{\text{minimize}} \quad \frac{1}{2}\|A(x^{(k)} + d) - b\|^2.$$

Its solution makes $x^{(k)} + d$ a least-squares solution. Therefore, with a full step, Gauss–Newton solves a linear least-squares problem in one iteration.

Gauss–Newton is exact when the residual functions are linear.

Possible difficulties

Gauss–Newton may encounter difficulties when:

- J_k does not have full column rank;
- the residuals remain large near the solution;
- the residual functions are strongly nonlinear;
- the starting point is far from a suitable solution;
- the full step does not reduce the objective;
- several local minimizers exist.

Possible remedies include:

- using a shorter step length;
- solving the linear least-squares subproblem with a rank-revealing method;
- regularizing the step;
- using a method that combines Gauss–Newton and gradient information.

Gauss–Newton exploits structure, but remains a local iterative method.

How the preceding lectures combine

From L10: linear least squares

$$\underset{d}{\text{minimize}} \quad \frac{1}{2}\|Ad - b\|^2.$$

From L11: iterative local models

$$x^{(k+1)} = x^{(k)} + t_k d^{(k)}.$$

Today: linearize nonlinear residuals

$$h(x^{(k)} + d) \approx h_k + J_k d.$$

Combining these ideas gives

$$\boxed{\underset{d}{\text{minimize}} \quad \frac{1}{2}\|J_k d + h_k\|^2.}$$

Gauss–Newton is a Newton-type method
built from linear least-squares subproblems.

Main ideas

After this lecture, you should be able to:

- formulate a nonlinear least-squares problem;
- derive the gradient and Hessian of a sum-of-squares objective;
- explain the Gauss–Newton Hessian approximation;
- linearize the residual vector;
- formulate the Gauss–Newton subproblem as a linear least-squares problem;
- derive the Gauss–Newton normal equations;
- perform a Gauss–Newton iteration in a small problem;
- explain when the Gauss–Newton approximation is expected to work well;
- describe situations in which step control or regularization is needed.

Linearize the residuals, solve a linear least-squares problem, and iterate.